

The v4 Dataset: Honest Evaluation Findings and the Shallow–Steep Coverage Gap

Audit results (June 9–11, 2026) and the design of `combined_dataset_v4`

June 11, 2026

Executive summary

A week of diagnostics established that most of the CS-DNO surrogate’s apparent long-rollout failures were artifacts of the measurement, not the model. After fixing the evaluation (f64 harness, truth-validity gating), the model has exactly *one* systematic weakness: **shallow water** ($h \lesssim 0.1$) **at moderate steepness** ($a/h \gtrsim 0.15$) **on non-soliton states**, a region where the training set has essentially no coverage. The v4 dataset adds a purpose-built *shallow–steep wavetrain* family ($\sim 480\text{k}$ samples) to the otherwise-unchanged v3 corpus, followed by a from-scratch retrain.

1 Findings that reshaped the evaluation

1.1 The f32 harness floor

All architectures (FNO, SpectralDNO, CS-DNO) plateaued at $\text{med} \approx 0.11 \text{ rel-}L^2(\eta)$ at $t = 200$ on Tanaka soliton rollouts. An oracle test — running the *analytic* order-6 operator through the same f32 surrogate rollout harness — hit the same plateau (0.112 vs. the model’s 0.1145), proving the plateau was float32 harness precision, not model error. The fix is `eval_suite --f64_harness` (integrator state and spectral ops in f64; the network still evaluates in f32). Truth self-convergence under substep doubling is $\sim 3 \times 10^{-9}$ at $t = 200$, so the f64 reference is converged.

Regime	f32 harness med	f64 harness med	ratio
tanaka_g0	0.1145	0.0420	$2.7\times$
tanaka_g1	~ 0.11	0.0443	$2.5\times$
bf_g0 / bf_g1 / bf_modal	0.09–0.14	0.0044 / 0.0067 / 0.0055	$20\text{--}26\times$

Table 1: Baseline CS-DNO ($t = 200$ median $\text{rel-}L^2(\eta)$). The Benjamin–Feir “problem” was almost entirely a harness artifact.

1.2 Truth-validity gating

Auditing every IC flagged as divergent (final $\text{rel} > 1$ or NaN) against the truth’s own health showed one regime’s entire divergence rate was a truth failure: on `bf_g1` IC15 the *truth integrator itself* goes non-finite at $t \approx 146$ with Hamiltonian drift 3.2×10^{-2} (healthy trajectories conserve to $\sim 10^{-7}$), while the surrogate stays finite. `eval_suite` now excludes ICs whose truth is non-finite or whose energy drift exceeds 10^{-3} from every aggregate (recorded as `truth_invalid_ics` in summaries).

Two caveats established alongside: (i) the saved-MATLAB cross-check (`saved_truth_reproduction`) is *not* a per-IC validity signal — two legitimate integrators decorrelate to $O(1)$ by late times on every bf IC, including ones the surrogate tracks to 0.4%; late-time pointwise error on chaotic regimes is shadowing-limited, and divergence-rate differences of ± 1 IC are noise. (ii) After gating, the only unambiguous tail failures are two isolated model NaNs in 80 coherent-regime ICs (`tanaka_g1` IC6 at $t \approx 45$; `bf_modal` IC4 at $t \approx 150$).

Regime	div. (raw)	div. (gated)	med (gated)	excluded ICs
<code>tanaka_g0</code>	6.2%	6.2%	0.0420	—
<code>tanaka_g1</code>	18.8%	18.8%	0.0443	—
<code>bf_g0</code>	0%	0%	0.0044	—
<code>bf_g1</code>	6.2%	0%	0.0067	IC15 (truth NaN)
<code>bf_modal</code>	6.2%	6.7%	0.0055	IC6 (truth drift 3.5×10^{-3})
<code>linear</code>	50%	27%	0.084	5 ICs (truth drift $> 10^{-3}$)

Table 2: Baseline CS-DNO, f64 harness, before/after truth-validity gating.

1.3 The “linear” regime is a shallow-steepness sweep

The eval regime named *linear* places all 16 ICs at a single shallow depth $h = 0.038$ (dominant mode $kh \approx 0.7$) with initial amplitude swept up to $a_0/h = 0.30$. Outcomes sort *monotonically by steepness*: every IC with $a_0/h \leq 0.12$ is clean (rel 0.003–0.105); the transition band 0.14–0.16 fails intermittently (one model NaN); everything at $a_0/h \geq 0.20$ fails — and at $a_0/h \geq 0.2$ the truth itself violates energy conservation on 5 of 16 ICs (drift 1.5×10^{-3} to 1.6), consistent with physical steepening/breaking beyond the solver’s validity. The surrogate’s failure signature on the surviving steep ICs is an over-amplified harmonic cascade: 1.9–3.3 $\times$ the truth’s spectral power at harmonics $2k$ – $4k$ of the fundamental. This is real shallow-water steepening physics that the model overdrives — the worst gated median on the board (0.084).

2 The one real gap, and why more solitons don’t fix it

After the corrections above, the surviving failure cluster is shallow water at moderate steepness. Three measurements pin the mechanism:

1. On exact soliton states — including the steepest whose rollouts die — the model’s single-step error is $\sim 2 \times 10^{-4}$: solitons are *well* learned (2M training rows, half below $h = 0.05$).
2. Below $h = 0.1$, v3 contains *only* soliton states. The **linear** training family stops at $h = 0.1$; the failing eval regime sits at $h = 0.038$.
3. A rollout’s own error continually creates near-soliton (but non-soliton) states; single-step accuracy there is the binding constraint, and at $h \lesssim 0.02$ on-manifold single-step error is already 10–70 $\times$ elevated.

Hence v4 adds generic shallow steep *wavetrains* (which also directly cover the failing “linear” regime), not more solitons.

3 The v4 dataset

3.1 New family: shallow–steep wavetrains (source 11)

Generator: `solver/gen_data/generate_shallow_steep_dataset.py`, mirroring the bf/tanaka production pipeline (f64 rollout, GL2 integrator, `filter_fraction=0.25`, order-6/pad-8 labels *with* the 0.25 lowpass).

ICs	1–5 cosine modes near a dominant mode; per-mode finite-depth linear ξ transfer with random $\pm$ direction signs
Depth	$h \sim \log \mathcal{U}[0.02, 0.12]$
Dominant mode	target $kh \sim \mathcal{U}[0.2, 1.5]$
Amplitude	$a/h \sim \mathcal{U}[0.03, 0.25]$, total steepness capped at $ka \leq 0.15$
Evolution	$t \in [0, 120]$, $dt = 0.08$, 120 snapshots/case (uniform)
Truth-health gate	reject case if Hamiltonian drift $> 10^{-5}$, any non-finite value, or $\min \eta \leq -0.9h$
Volume	480,000 samples (two 240k shards, one per GPU)

Table 3: Shallow–steep family specification.

Validation on the smoke batch: stored labels match a filtered f64 recompute to median 9×10^{-8} ; kept-case energy drift median $\sim 10^{-8}$ (three orders inside tolerance); rejection rate 5–9%, concentrated at nominal $a/h \gtrsim 0.18$ — precisely where the truth is known to fail; evolution naturally steepens states up to $\max |\eta|/h \approx 0.5$ with energy still conserved, providing the off-manifold coverage the rollouts need. All amplitudes sit well inside v3’s normalization envelope.

3.2 Assembly and retrain

`combined_dataset_v4.npz` = v3 byte-identical (5,993,989 rows; no relabeling, no resampling) + the new family appended as source 11 ($\sim 7.4\%$ of the corpus). No global reshuffle is needed: the trainer’s `build_split_indices` applies a seeded random permutation, scattering new rows uniformly over train/val/test. Normalization statistics are recomputed from v4 (no sidecar pinning — this is a from-scratch train, not a resume).

Retrain: CS-DNO, identical hyperparameters to the v3 baseline (modes 64, width 512, 8 blocks, latent 256, `cs_mult_hidden` 128, Sobolev-1 loss, batch 256, lr 2×10^{-4} , wd 10^{-4} , 40 epochs), 2 GPUs.

3.3 Success criteria (f64 harness, truth-gated)

- **linear** regime: gated divergence $< 27\%$ and median < 0.084 (primary target — this is the covered gap);
- tanaka medians hold: $\leq 0.042 / 0.044$;
- bf medians within $\sim 20\%$ of $0.0044/0.0067/0.0055$; `random_sea/stokes` unchanged;
- stretch: the two isolated model NaNs disappear.

Out of scope by design: chaotic-IC pointwise tails (shadowing-limited; report invariants instead) and the truth-invalid ICs (excluded by the gate).